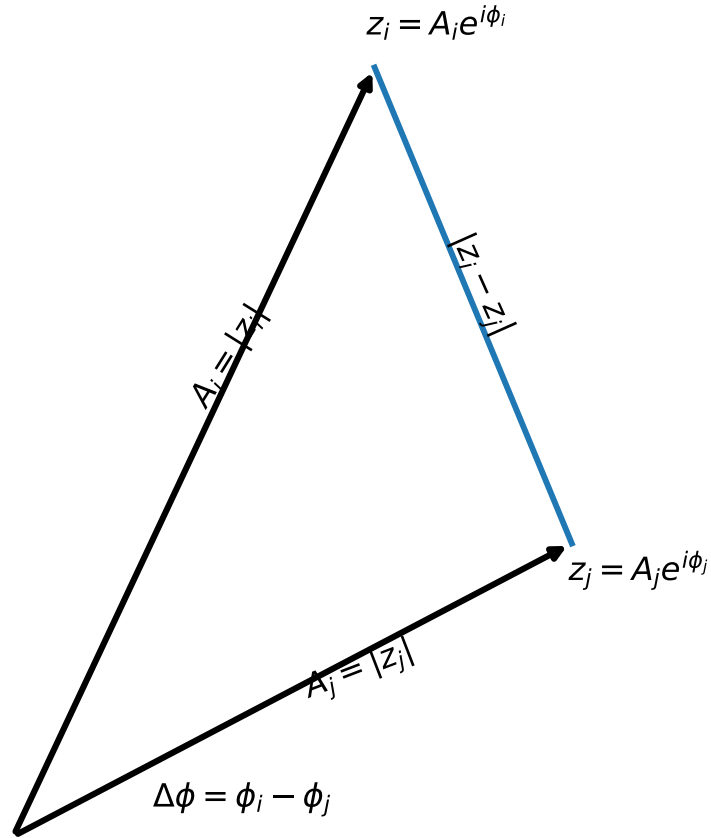


A Complex-coordinate geometry



B Two exact decompositions and one diagnostic

$$D_{\text{FULL}} = D_{\text{AMP}} + D_{\text{AP}}$$

$$D_{\text{FULL}} = D_{\text{ENERGY}} + D_{\text{INTERFERENCE}}$$

$$D_{\text{AMP}} = D_{\text{ENERGY}} - D_{\text{PRODUCT}}$$

$$D_{\text{AP}} = D_{\text{PRODUCT}} + D_{\text{INTERFERENCE}}$$

$$D_{\text{ENERGY}} = \sum_m (A_{i,m}^2 + A_{j,m}^2)$$

$$D_{\text{PRODUCT}} = \sum_m 2A_{i,m}A_{j,m}$$

$$D_{\text{INTERFERENCE}} = -\sum_m 2A_{i,m}A_{j,m}\cos(\Delta\phi_m)$$

$$D_{\text{ANGLE}} = \sum_m 2[1 - \cos(\Delta\phi_m)]$$

PRODUCT: co-activation magnitude; independent of phase

INTERFERENCE: signed phase cross-term; aligned phases reduce FULL distance

ANGLE: amplitude-free phase diagnostic (not an exact FULL component)